ENGINEERING TECHNICAL WHITEPAPER

# Reproducing the Ablution-Block Orthographic Drawing

# Third-Angle Projection at 1:50 on an A3 Sheet

Dimensioning, scale calculation, sheet layout, the 45° mitre-line method, and a full schedule of measured and assumed dimensions — prepared as a working reference for hand or CAD reproduction of Figure 1 (Individual Assignment 2, Q4).

| | |
|---|---|
| **Subject** | Engineering Drawing / Orthographic Projection |
| **Deliverable** | A3 drawing: Plan + Front & End elevations, third-angle, scale 1:50, with title block |
| **Source** | Figure 1 – Ablution block orthographic view (image source: Tantallon Estates) |
| **Standards** | ISO 216 (paper), ISO 5457 (drawing sheets), ISO 5456-2 / ASME Y14.3 (projection) |
| **Status** | Prepared for KARL • Civil Engineering • reproduction workflow |

# 1 Purpose and scope

This whitepaper confirms every measurement needed to reproduce the ablution-block drawing (Figure 1) on a single A3 sheet using **third-angle orthographic projection** at a scale of **1:50**. It sets out (a) the correct A3 sheet size and drawing frame from the governing paper standards; (b) the arithmetic that converts every real dimension to its length on paper at 1:50; (c) how the three views are arranged and how depths are transferred between them with a 45° mitre line; (d) a complete schedule of the dimensions that are printed on the source drawing; and (e) a schedule of the dimensions that are *not* shown — principally vertical heights — with assumed values fixed by ordinary building-engineering baselines and clearly flagged as assumptions.

The source shows three related views: a long **front elevation**, a short **end elevation** that reveals the mono-pitch roof, and a fully dimensioned **plan** of a male/female ablution block with WCs, urinals and wash-hand basins, drained by a Ø110 PVC pipe to a septic tank and soak-away. All plan dimensions are given; the heights are read off by scale or assumed.

# 2 Sheet specification — A3 and the drawing frame

The drawing is produced on a trimmed **A3** sheet. Under the ISO 216 A-series the sides are in the ratio 1 : 1.414 (one to the square root of two), so that halving a sheet gives the next size down at the same proportions.

A3 measures **297 × 420 mm** (i.e. 420 mm long × 297 mm high in landscape). It is exactly half of A2 and twice A4, and is the standard size for detailed single-sheet building details.

| Sheet | Trimmed size (mm) | Drawing space, ISO 5457 (mm) | Relation |
|---|---|---|---|
| A2 | 420 × 594 | 400 × 564 | 2 × A3 |
| **A3** | **297 × 420** | **277 × 390** | **this sheet** |
| A4 | 210 × 297 | 180 × 277 | ½ × A3 |

**Drawing frame & margins (ISO 5457).** Inside the trimmed edge a border frame is drawn leaving a **20 mm** filing margin on the left (binding) edge and **10 mm** on the other three edges. For A3 this yields a usable drawing space of **390 × 277 mm**. A **title block** is placed in the bottom-right corner (typically up to 180 mm wide). Everything — the three views, the mitre construction and the title block — must sit inside the 390 × 277 frame.

# 3 Scale 1:50 — the conversion

A scale of **1:50** means **1 mm on paper = 50 mm in reality**; every real dimension is divided by 50 to get its plotted length. Equivalently 1 m of building plots as 20 mm, and 20 mm on the sheet represents 1 m. The table below converts the principal dimensions; apply the same rule (÷ 50) to any dimension not listed.

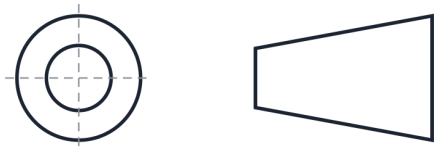
| Element | Real (mm) | On A3 at 1:50 (mm) | Element | Real (mm) | On A3 at 1:50 (mm) |
|---|---|---|---|---|---|
| Overall length | 8575 | **171.5** | WC depth 2115 | 2115 | **42.3** |
| Overall depth | 5045 | **100.9** | Wall thickness 115 | 115 | **2.3** |
| End bay 2615 | 2615 | **52.3** | Floor level +300 | 300 | **6** |
| Back zone depth 2000 | 2000 | **40** | Wall height 2400 (assumed) | 2400 | **48** |
| Central bay 1885 | 1885 | **37.7** | High eave 2841 (assumed) | 2841 | **56.82** |
| Mid zone 1585 | 1585 | **31.7** | Roof rise 441 (assumed) | 441 | **8.82** |
| Entrance 1500 | 1500 | **30** | Octagon vent 420 (assumed) | 420 | **8.4** |
| WC cubicle 1000 | 1000 | **20** | | | |

**Fit check.** At 1:50 the front elevation plots 171.5 mm wide and about 66 mm high; the plan 171.5 × 100.9 mm; the end elevation 100.9 × 66 mm. Arranged as below with ~30–40 mm gaps, the group needs roughly **312 × 202 mm** — comfortably inside the 390 × 277 mm A3 frame, leaving room for the title block and mitre construction. **1:50 on A3 is therefore confirmed as appropriate.**

## 4 Third-angle projection — arrangement of views

In third-angle projection the object sits in an imaginary glass box and each face is projected outward onto the nearest pane; unfolding the box gives the standard layout. The rule is simple and intuitive: **the view you take looking down goes above; the view looking from the right goes to the right.** So for this block:

- **Front elevation** — the principal view (longest face, most information): centre-left.
- **Plan (top view)** — placed **directly above** the front view, vertically aligned (shares the same length).
- **End (right-side) elevation** — placed **to the right** of the front view, horizontally aligned (shares the same heights).

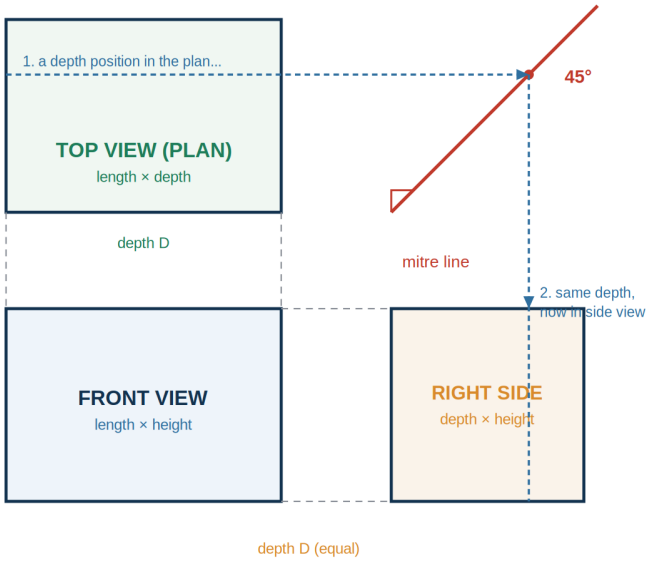



*Third-angle projection symbol — place it in the title block. The frustum's narrow end points toward the circular (end) view; the right view falls to the right of the front.*

Because adjacent views share dimensions, features line up: heights are common to the front and end elevations (project horizontally between them), and the length is common to the front view and the plan (project vertically between them). Only the **depth** is not directly shared between the plan and the end elevation — and that is exactly what the mitre line solves.

## 5 The 45° mitre-line method — mapping the views to each other

Depth appears **vertically** in the plan but **horizontally** in the end elevation. A 45° mitre line converts one into the other so the two views stay in perfect register. This is the “map the elevations over 45° to each other” step:

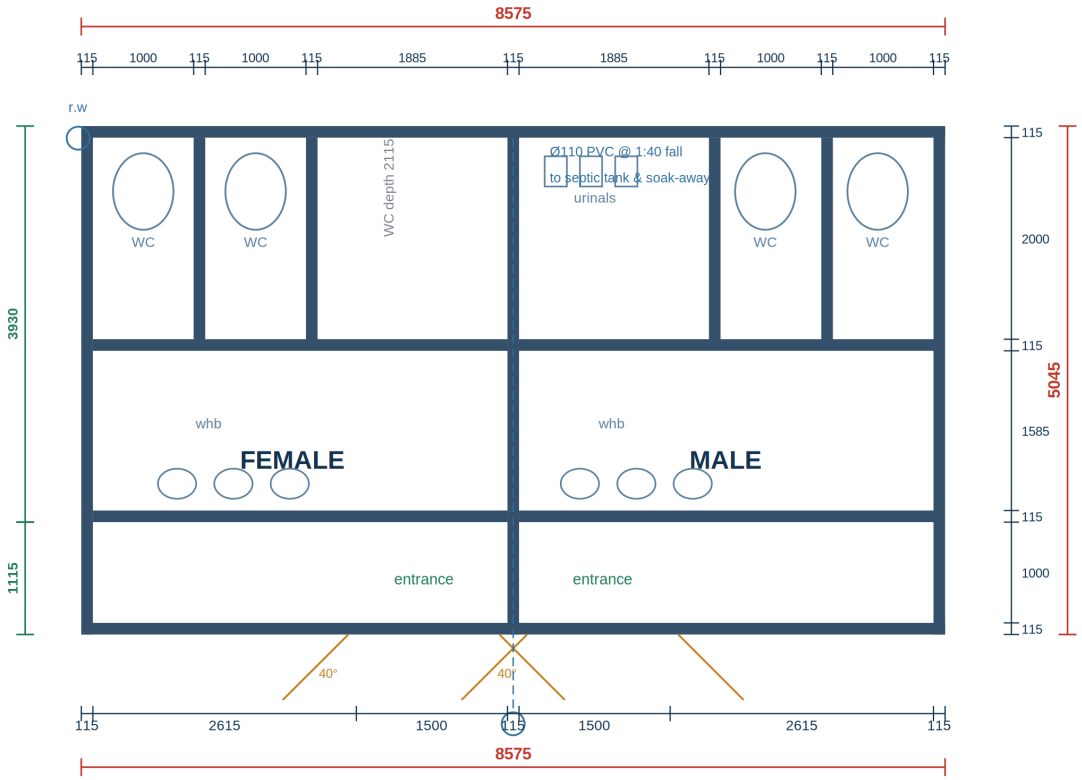



*Third-angle view arrangement with the 45° mitre line transferring depth from the plan into the end elevation. Depth D is identical in both views.*

| Step | Action |
|---|---|
| **1. Draw the front view** | Set the front elevation in the lower-left of the group, working to the 1:50 plotted lengths and to the assumed heights (Section 7). |
| **2. Project the plan up** | From every important point on the front view, draw **light vertical projectors** upward; the plan is built on these, directly above the front view, so lengths agree automatically. |
| **3. Project heights across** | From every height on the front view draw **light horizontal projectors** to the right; the end elevation is built on these, so all heights agree. |
| **4. Draw the mitre line** | At the top-right, where a horizontal from the plan's front edge meets a vertical from the end view's front edge, draw a line at exactly **45°**. |
| **5. Transfer the depths** | For each depth feature in the plan, run a horizontal projector right to the mitre line, then **turn 90° downward** into the end elevation. The mitre swaps the vertical plan-depth into the horizontal elevation-depth, point for point. |
| **6. Firm up & annotate** | Line in the visible outlines (bold, ~0.5–0.7 mm), keep projectors light, add dimensions, level tags, notes, hatching and the title block. |

# 6 Measured dimensions (given on the drawing)

These are read directly from Figure 1 and are authoritative — reproduce them exactly. All values in millimetres. The dimension chains close on the overall sizes (checked below).




*Key plan re-drawn to the given dimension chains (schematic, not to final scale). Female/male mirror about the central 115 mm wall.*

| Run / chain | Segments (mm, in order) | Sum |
|---|---|---|
| **Length – top chain** | 115 · 1000 · 115 · 1000 · 115 · 1885 · 115 · 1885 · 115 · 1000 · 115 · 1000 · 115 | **8575** |
| **Length – bottom chain** | 115 · 2615 · 1500 · 115 · 1500 · 2615 · 115 | **8575** |
| **Depth chain (right)** | 115 · 2000 · 115 · 1585 · 115 · 1000 · 115 | **5045** |
| **Depth sub-totals (left)** | 3930 (back+mid) · 1115 (front) | **5045** |

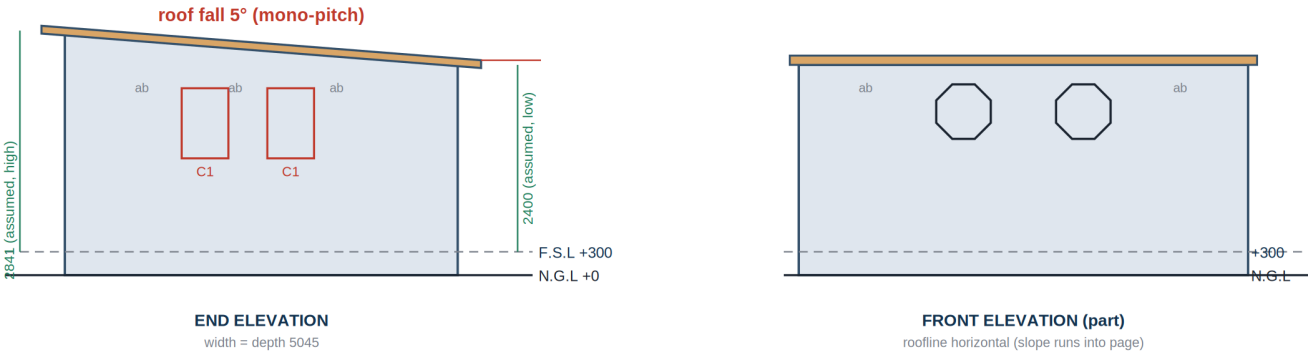
| Item | Value (mm) | Notes |
|---|---|---|
| **Overall length × depth** | 8575 × 5045 | External, over 115 walls |
| **Wall thickness** | 115 | Single-skin throughout (incl. central divider) |
| **WC cubicle** | 1000 wide, 2115 deep | 4 off (2 female, 2 male) |
| **Central bays** | 1885 wide | urinal / circulation zones |
| **Entrances** | 1500 wide each | 2 off, either side of central wall |
| **Finished floor level** | +300 above NGL | F.S.L +300 / N.G.L +0 |
| **Drainage** | Ø110 PVC @ 1:40 | fall to septic tank & soak-away |
| **Fixtures** | — | 4 WC, urinals, 6 whb, air-bricks (ab) at high level |

# 7 Assumed dimensions (engineering baselines)

The elevations are **not vertically dimensioned** on the source. The values below are assumed from standard building practice and from proportions measured off the drawing at scale. **They must be labelled "ASSUMED" on the sheet.** If a project specification is available, it overrides these.

| Item | Assumed value | Basis / engineering judgement |
|---|---|---|
| **Wall height, low eave (FFL→wallplate)** | 2400 mm | Standard clear height for ablutions; gives comfortable headroom |
| **Roof pitch** | 5° mono-pitch | **Measured off the end elevation (5.0°)**; also the practical min. fall for sheeting |
| **Roof rise over 5045 depth** | 441 mm | 5045 × tan 5°; sets high eave = 2400 + 441 ≈ 2841 mm |
| **Eaves overhang** | 300 mm | Typical for weather protection to masonry |
| **Octagonal vents (front)** | ≈420 mm across | **Measured off drawing**; decorative ventilation openings, ~1800 mm to centre |
| **Windows C1 (end elev.)** | 600 × 900 h | Small high-sill steel window; sill ~1200 mm for privacy |
| **Door leaf at entrances** | 900 × 2100 h | Standard external door size |
| **Air bricks (ab)** | 215 × 140 | Standard vent brick; placed at high level ~2100 mm |
| **Entrance ramp splay** | 40° wings | **Given on plan ("Angle 40°")**; ramp gradient 1:12 for access |
| **Strip foundation** | 600 × 230 | Typical for single-storey masonry; top ≥200 below NGL (for section only) |
| **Floor slab** | 100 mm | Concrete on 250µm DPM on compacted fill (for section only) |

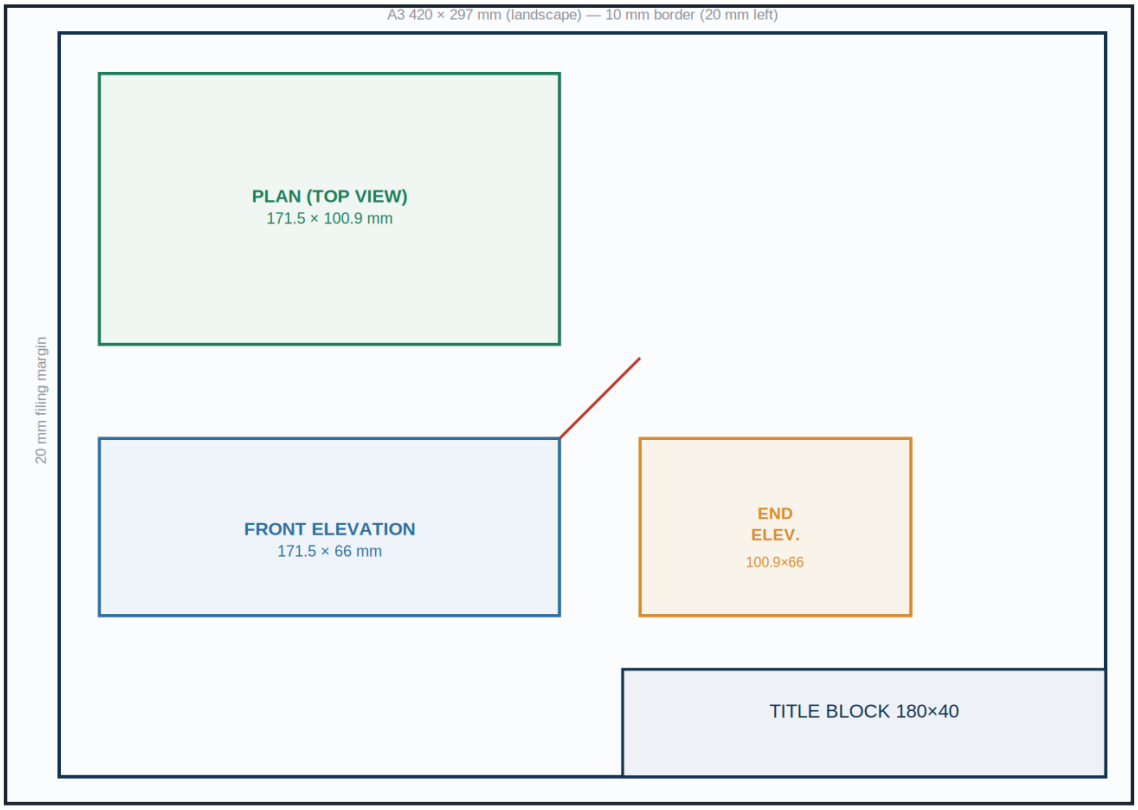
# 8 Angles and circular features




*Elevation scheme — 5° mono-pitch roof, level tags, C1 windows and octagonal vents. Heights marked “assumed” per Section 7.*

| Feature | Value | Source | How to set it out |
|---|---|---|---|
| **Roof pitch** | 5° | Measured | Fall across the 5045 depth in the end elevation; rise 441 mm |
| **Ramp splay** | 40° | Given (plan) | Set the ramp cheek wings at 40° to the entrance line |
| **Octagon vents** | ≈Ø420 | Measured | Regular octagon; two, centred in the front-elevation middle bay |
| **Rain-water point (r.w)** | — | Given (plan) | Gully symbol top-left; start of the Ø110 run |
| **Inspection/gully** | — | Given (plan) | Circle on the pipe centre-line at the entrance; Ø110 @ 1:40 |

# 9 Sheet layout — where each view sits on the A3




*Recommended A3 landscape layout (third-angle). Coordinates below are in millimetres from the bottom-left corner of the trimmed sheet.*

| Element | Bottom-left (x, y) | Size w × h (mm) |
|---|---|---|
| **Trimmed sheet** | 0, 0 | 420 × 297 |
| **Border frame** | 20, 10 | 390 × 277 (20 left / 10 others) |
| **Plan (top view)** | 35, 171 | 171.5 × 100.9 |
| **Front elevation** | 35, 70 | 171.5 × 66 |
| **End elevation** | 236.5, 70 | 100.9 × 66 |
| **Mitre construction** | from ~206, 136 | 45° into top-right gap |
| **Title block** | 230, 10 | 180 × 40 |

Gaps between views (~30–35 mm) leave room for dimension lines and the mitre; nudge the group left/up if you enlarge the title block.

## 10 Step-by-step drawing procedure

| # | Stage | What to do |
|---|---|---|
| 1 | **Prepare the sheet** | A3 landscape; draw the border (20 mm left, 10 mm elsewhere) and block out the title block bottom-right. |
| 2 | **Set out the plan** | Plot the length chain (8575 → 171.5 mm) and depth chain (5045 → 100.9 mm) at 1:50; add internal walls (each 115 → 2.3 mm), cubicles, central divider and openings. |
| 3 | **Project the front elevation** | Drop vertical projectors from the plan to build the front elevation below it; set floor at F.S.L +300 and wall to the assumed heights. |
| 4 | **Set up the mitre** | Draw the 45° line top-right; project heights right from the front view and depths up-then-over from the plan to build the end elevation. |
| 5 | **Add roof & features** | 5° mono-pitch (rise 441 over the depth), octagon vents, C1 windows, air-bricks, ramps at 40°, drainage run Ø110 @ 1:40. |
| 6 | **Dimension & annotate** | Repeat the given chains; tag F.S.L/N.G.L; note “wall plaster / painted to client's specification”; mark every assumed value “ASSUMED”. |
| 7 | **Title block & symbol** | Complete the title block (Section 11) and add the third-angle projection symbol. |

## 11 Title block content

The title block (bottom-right) should carry at least the following, plus the third-angle symbol and a scale bar:

| Field | Entry | Field | Entry |
|---|---|---|---|
| **Project** | Ablution Block | **Scale** | 1:50 |
| **Drawing title** | Plan & Elevations | **Sheet size** | A3 |
| **Drawn by** | K. (student no.) | **Projection** | Third angle |
| **Date** | — | **Units** | mm |
| **Checked** | — | **Dwg no.** | A2-Q4 |

## 12 References


- ISO 216 — Writing paper and certain classes of printed matter — Trimmed sizes (A, B, C series). A3 = 297 × 420 mm.
- ISO 5457 — Technical product documentation — Sizes and layout of drawing sheets (frame, margins, title-block zone).
- ISO 5456-2 / ASME Y14.3 — Projection methods and the first/third-angle projection symbols.
- Engineering ToolBox — Paper drafting sizes, ISO 216 A/B/C series (A3 drawing space 277 × 390 mm).
- Standard orthographic-projection texts — the 45° mitre-line method for transferring depths between the plan and the end view.


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Note on assumptions: values marked “assumed” are engineering baselines for a single-storey masonry ablution block and are provided so the sheet can be completed where the source omits vertical dimensions. Confirm against any project specification before construction use.